

Daily Geometry Drill #7 — Answers & Investigations

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

Section A Rapid Recognition

1. **Answer:** "a circle of radius 5 centred at the origin"

Method: $\sqrt{x^2 + y^2} = 5$ rearranges to $x^2 + y^2 = 25$, a circle of radius 5.

Investigate further: Forgetting to square: quoting radius 25 or a square.

2. **Answer:** $x = 2$

Method: Squaring $x^2 + y^2 = (x - 4)^2 + y^2$: $x^2 = (x - 4)^2$, so $x = 2$.

Investigate further: Giving the single midpoint $(2, 0)$ instead of the whole vertical line.

3. **Answer:** 4

Method: On the x -axis, between A and B we have $PA = x$ and $PB = 6 - x$, so $x = 2(6 - x)$, giving $x = 4$. The Apollonius circle's two x -axis endpoints are $x = 4$ and $x = 12$.

Investigate further: Choosing $x = 2$ (halving 6) or the external endpoint $x = 12$.

4. **Answer:** $y = 1$ or $y = 7$

Method: $|y - 4| = 3$ gives $y = 1$ and $y = 7$, two parallel lines 3 units from $y = 4$.

Investigate further: Keeping only one line $y = 1$ and missing its mirror $y = 7$.

5. **Answer:** $(2, 1)$

Method: A point equidistant from focus $(0, 1)$ and directrix $y = -1$ satisfies $\sqrt{x^2 + (y - 1)^2} = |y + 1|$, i.e. $x^2 = 4y$; $(2, 1)$ works since $4 = 4$.

Investigate further: Naming a point on the y -axis (which only hits the vertex origin for integer y) or mislaying the focus/directrix distance.

6. **Answer:** $(x - 1)^2 + (y - 2)^2 = 9$

Method: Points at distance exactly 3 from $(1, 2)$ form the circle centred there with radius 3.

Investigate further: Writing distance = 3 unsquared, or using the point as an offset in the wrong sign.

7. **Answer:** "a circle of radius 1 centred at the origin"

Method: Q runs over $x^2 + y^2 = 4$ (radius 2); the midpoint $M = OQ/2$ scales radii by $\frac{1}{2}$, giving

$$x^2 + y^2 = 1.$$

Investigate further: Halving the equation's radius linearly to $\sqrt{2}$ or keeping the original radius 2.

8. **Answer:** 1

Method: $x^2 + y^2 = 2x$ rearranges to $(x - 1)^2 + y^2 = 1$, radius 1.

Investigate further: Answering $\sqrt{2}$ (reading the constant 2 as r^2 before completing the square).

9. **Answer:** false

Method: The point $(\sqrt{2}, \sqrt{2})$ lies on $x^2 + y^2 = 4$ yet has $x \neq \pm 2$ and $y \neq \pm 2$; the implication fails in both coordinates.

Investigate further: Mistaking the circle's bounding box for its points; only the four axis points hit ± 2 .

10. **Answer:** $x = 3$

Method: A and B are at equal height $y = 3$, so the perpendicular bisector is the vertical line $x = \frac{1+5}{2} = 3$.

Investigate further: Writing the horizontal line $y = 3$ through both points.

Section B Manipulation Drills

1. **Answer:** A) 4

Method: On the x -axis, $|x| = 2|x - 6|$ gives the endpoints $x = 4$ and $x = 12$; the circle between them has centre $(8, 0)$ and radius 4.

Investigate further: Quoting the centre distance 8 (option B) or the gap 6 (option D).

2. **Answer:** B) $y = 2$

Method: $x^2 + y^2 = x^2 + (y - 4)^2$ gives $y = 2$, the horizontal bisector.

Investigate further: Answering $x = 2$ (wrong orientation) or $y = 4$.

3. **Answer:** B) 4π

Method: The disk has radius 2, so its area is $\pi r^2 = 4\pi$.

Investigate further: Using $\pi r = 2\pi$ (option A) or $r^2 = 16$ (option D).

4. **Answer:** 2π

Method: Circumference $= 2\pi r = 2\pi \cdot 1 = 2\pi$.

Investigate further: Mixing circumference with area (π) or doubling the radius to 4π .

5. **Answer:** C) $x^2 + y^2 < 2$

Method: If $x^2 + y^2 < 1$ then certainly $x^2 + y^2 < 2$; option A is not necessary (a point with $x^2 + y^2 = 0.75$ is inside but not $< \frac{1}{2}$).

Investigate further: Picking a circle whose radius is smaller than the disk (option A) as if it were implied.

6. **Answer:** A) 3

Method: On the x -axis $|x| = 3|x - 2|$ gives internal $x = \frac{3}{2}$ and external $x = 3$; the right endpoint is 3.

Investigate further: Choosing the left endpoint $\frac{3}{2}$ or the y -absent centre guess.

7. **Answer:** B) $x = -\frac{1}{2}$ and $y = \frac{\sqrt{3}}{2}$

Method: This point satisfies $x^2 + y^2 = \frac{1}{4} + \frac{3}{4} = 1$ exactly, so it pins P on the circle. Option A puts P strictly *inside*; C and D do not force $x^2 + y^2 = 1$.

Investigate further: Choosing a strictly-inside condition (option A) because it is a *necessary* symptom of being on the circle.

8. **Answer:** 3

Method: Distance from $(3, 0)$ to the origin is $\sqrt{9} = 3$.

Investigate further: Answering 6 (the segment AB) or 9 (the squared distance).

9. **Answer:** $(2, 3)$

Method: In $x^2 + y^2 + 2gx + 2fy + c = 0$, the centre is $(-g, -f)$; with $2g = -4$ and $2f = -6$ we get $(2, 3)$.

Investigate further: Halving the wrong sign: $(4, 6)$ or negating incorrectly.

10. **Answer:** 2

Method: Completing the square: $(x - 2)^2 + (y + 3)^2 = (-9) + 4 + 9 = 4$, radius 2.

Investigate further: Quoting the constant 9 as the radius or r^2 before shifting.

Section C Substitution & Structure

1. **Answer:** A) 3

Method: The distance from the line $x = 3$ to $B = (6, 0)$ is $|6 - 3| = 3$.

Investigate further: Answering 6 (the gap $A-B$, option B).

2. **Answer:** A) $(8, 0)$

Method: Endpoints of the locus on the x -axis were $x = 4$ and $x = 12$, so the centre is their midpoint $(8, 0)$.

Investigate further: Choosing the B -side endpoint $(12, 0)$ or the midpoint of AB , $(3, 0)$.

3. **Answer:** A) $(2, 2)$

Method: $(2, 2)$ has $x^2 + y^2 = 8 < 9$; the others give 18, 13, 16.

Investigate further: Picking a boundary or an exterior point.

4. **Answer:** A) $x = 3$

Method: Equal ratios are the perpendicular bisector of AB , the line $x = 3$.

Investigate further: Answering the line through B (option C) or a diagonal.

5. **Answer:** B) 4

Method: Each of $(1, 2)$, $(2, 1)$, $(\sqrt{5}, 0)$, $(1, -2)$ satisfies $x^2 + y^2 = 5$.

Investigate further: Counting only the obvious axis pair or miscounting by sign.

6. **Answer:** B) 8

Method: The major axis equals the constant sum $PA + PB = 8$.

Investigate further: Quoting the focal distance 6 (option C) or the semi-major axis 4 (option A).

7. **Answer:** A) $a^2 + b^2 = 1$

Method: Being on the unit circle is exactly $a^2 + b^2 = 1$: necessary (only these points qualify) and sufficient (every such point is on it).

Investigate further: Choosing a merely-necessary condition (option B) or a merely-sufficient one.

8. **Answer:** C) 0

Method: $(x - 2)^2 + y^2 = 0$, a single point at $(2, 0)$: “radius” 0.

Investigate further: Quoting $\sqrt{4} = 2$ (option B) without completing the square.

Section D Challenge Ramp

1. **Answer:** A) 8

Method: Endpoints at $x = 4$ and $x = 12$ (from $|x| = 2|x - 6|$), so the chord between them on the x -axis is $12 - 4 = 8$.

Investigate further: Quoting the radius 4 (option B) or the centre-to- B distance.

2. **Answer:** A) 1

Method: $(x - 2)^2 + (y - 3)^2 = 12 - 4 - 9 + \text{shift} = 1$, radius 1.

Investigate further: Failing to subtract the completed-square constant; $\sqrt{12}$ (option D) is a snare.

3. **Answer:** A) $x^2 + y^2 < \frac{1}{2}$

Method: This forces $x^2 + y^2 < 1$ strictly, i.e. strictly inside the unit disk. Option C allows points outside the unit disk (e.g. $x^2 + y^2 = 1.5$).

Investigate further: Choosing the weaker disk $x^2 + y^2 < 2$ (which is not sufficient).

4. **Answer:** A)

Method: Expanding $x^2 + y^2 = 9((x - 2)^2 + y^2)$ yields $(x - \frac{9}{4})^2 + y^2 = \frac{9}{16}$, a circle centred at $(\frac{9}{4}, 0)$ of radius $\frac{3}{4}$ — not $(x - 2)^2 + y^2 = 9$. Fil's circle is centred at B and ignores A entirely.

Investigate further: Believing $(x - 2)^2 + y^2 = 9$ (option D) or claiming the ratio locus is a line (option C).

5. **Answer:** $x^2 + y^2 = 5$

Method: Through $(1, 2)$ with centre O : $r^2 = 1^2 + 2^2 = 5$.

Investigate further: Writing $(x - 1)^2 + (y - 2)^2 = 5$ (putting the given point as the centre).

Top 5 Patterns

- **A fixed-distance locus is a circle; square it to find the equation.** Distance = r means $x^2 + y^2 = r^2$; an offset centre shifts the squares. (see A1, A6, A8, B3, B4, D5)
- $\frac{PA}{PB} = k$ is an **Apollonius circle** when $k \neq 1$. Square the ratio, collect x, y , complete the square; endpoints on the x -axis come from $|x| = k|x - b|$. (see A3, B1, B6, C2, D1, D4)
- **Equal ratios ($k = 1$) are the perpendicular bisector.** The line $\frac{PA}{PB} = 1$ is the mirror line of the segment. (see A2, A10, B2, B8, C1, C4)
- **Distinguish *necessary* from *sufficient*.** A condition must be true for the claim (necessary) versus enough to force the claim (sufficient). (see B5, B7, C7, D3)
- **A line and a circle can both be “3 units away”; name both.** Absolute-value loci split into symmetric branches. (see A4, A9)

Common Traps

- **Reading a distance-equality as a point instead of a family.** $PA = PB$ names a whole line, not just the midpoint. (see A2, A10, C4)
- **Quoting the radius from r^2 before completing the square.** The constant in $x^2 + y^2 + 2gx + 2fy + c = 0$ is not r^2 ; the centred shift supplies it. (see B10, C8, D2, A8)
- **Forgetting the second solution when a modulus meets.** $|x| = k|x - b|$ has an internal and an external endpoint; a flagged ratio answer usually picks the wrong one. (see A3, B6, D1)
- **Confusing necessary with sufficient.** “Must be inside” and “is forced on the circle” are different strengths; one extra decimal of the radius flips the verdict. (see B5, B7, C7, D3)
- **Believing a claimed Apollonius equation without testing it.** Expand the squared ratio yourself; a circle centred at one focus is the classic decoy. (see D4)

“A locus is a sentence the entire family agrees on; a geometric claim is only as good as the condition you actually invoked.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths